

A to Z process bohot simple hai. Aapka **Django backend** ek bridge (pula) ki tarah kaam karega jo aapki website (Frontend) se file lega aur use aapke computer mein chalne wale **Ollama AI** ko bhej dega.

Yahan mukammal step-by-step guide hai:

---

### Step 1: Ollama ko "External Connections" ke liye tayyar karna

By default, Ollama sirf terminal se baat karta hai. Aapko use batana hoga ke Django backend bhi us se baat kar sakta hai.

1. Apne Windows mein "**Edit the system environment variables**" search karen.
  2. **Environment Variables** par click karen.
  3. "User variables" mein **New** par click karen:
    - **Variable name:** OLLAMA\_ORIGINS
    - **Variable value:** \*
  4. Ollama ko taskbar se **Quit** karen aur dobara restart karen.
- 

### Step 2: Django Backend mein "Brain" Setup karna

Aapko Django mein do cheezon ki zaroorat hogi: ek PDF se text nikalne ke liye aur dusra AI se baat karne ke liye.

1. Apne backend folder mein ye libraries install karen:

Bash

```
pip install ollama pymupdf
```

2. api/views.py mein ye function likhen jo file upload hote hi AI ko trigger karega:

Python

```
import fitz # PyMuPDF
```

```
import ollama
```

```
from django.http import JsonResponse
```

```
def generate_ai_quiz(request):
```

```
    if request.method == 'POST' and request.FILES.get('pdf_file'):
```

# 1. PDF File uthao

```
pdf_file = request.FILES['pdf_file']
```

# 2. Text Extract karo

```
doc = fitz.open(stream=pdf_file.read(), filetype="pdf")
```

```
text = ""
```

```
for page in doc:
```

```
    text += page.get_text()
```

# 3. Local Llama 3.1 ko bhenjo (localhost:11434 par)

```
response = ollama.chat(model='llama3.1:8b', messages=[
```

```
{
```

```
    'role': 'user',
```

```
    'content': f"Generate 5 MCQs in JSON format from this text: {text[:4000]}",
```

```
},
```

```
], format='json')
```

# 4. Result wapis frontend ko bhenjo

```
return JsonResponse(response['message']['content'], safe=False)
```

---

### Step 3: Frontend (React) ko Connect karna

Aapka jo **NewTraining.jsx** page hai, usme jab "Save" ya "Create" button dabega, to wo is Django URL ko hit karega.

1. NewTraining.jsx mein FormData ka istemal karen taake PDF file backend tak pahunch sake.
2. Backend jab MCQs wapis karega, to use state mein save karke screen par dikha den.

---

### Step 4: Video ke liye AI (Whisper)

Agar aap video se quiz banana chahte hain, to process same rahega bas PDF ki jagah aapko **Whisper** library use karni hogi jo video ki awaz ko text mein badal degi, phir wo text Llama ko jayega.

---

Summary Table: Connection Flow

| Step | Component | Action                                            |
|------|-----------|---------------------------------------------------|
| 1    | Frontend  | Admin PDF upload karta hai.                       |
| 2    | Backend   | Django PDF se text nikalta hai.                   |
| 3    | AI Engine | Ollama (Llama 3.1) text ko quiz mein badalta hai. |
| 4    | Database  | Django quiz ko db.sqlite3 mein save karta hai.    |

Kya aap chahte hain ke mein aapko woh exact React code likh kar doon jo aapki "NewTraining.jsx" mein lagega taake file backend tak sahi tarah jaye?

Ab scenario thora badal gaya hai. Jab AI (Ollama) aik computer par ho aur Development (Django/React) dusre computer par, to aapko **Networking** ka istemal karna parega. Yahan A to Z process hai ke aap "AI wale PC" ko "Development wale PC" se kaise connect karenge:

---

1. AI Wale PC ki IP Address Nikalna

Jis computer par **Llama 3.1** install hai, wahan ye kaam karen:

- 1. **CMD** kholen aur ipconfig likhen.
  - 2. **IPv4 Address** note kar len (Maaloom karen ke ye 192.168.x.x jaisa ho).
    - Farz karen AI PC ki IP hai: 192.168.10.20
-

## 2. AI PC par Ollama ko "Open" karna

Ollama default taur par dusre computers ki requests block karta hai. Isay har jagah se access karne ke liye ye steps lazmi hain:

1. **Environment Variables** mein jayen (Windows Search mein "env" likhen).
  2. **System Variables** mein New par click karen:
    - **Variable Name:** OLLAMA\_HOST
    - **Variable Value:** 0.0.0.0
  3. Aik aur **New** variable banayen (CORS allow karne ke liye):
    - **Variable Name:** OLLAMA\_ORIGINS
    - **Variable Value:** \*
  4. Ollama ko Taskbar (System Tray) se **Quit** karen aur dobara chalayen.
- 

## 3. Connection Test Karen (Development PC se)

Ab apne dusre PC (jis par aap coding kar rahe hain) par jayen aur check karen ke kya wo AI PC ko "dekh" sakta hai:

1. Browser kholen aur URL mein likhen: `http://192.168.10.20:11434` (Apni nikali hui IP use karen).
  2. Agar screen par "**Ollama is running**" likha aa jaye, to samjhen connection kamyab hai!
- 

## 4. Django Backend mein IP Change Karen

Ab aapko apne Django code mein localhost ki jagah AI PC ki IP likhni hogi.

**api/views.py** mein tabdeeli:

Python

```
import ollama
```

```
# Connection setup
```

```
client = ollama.Client(host='http://192.168.10.20:11434') # Yahan AI PC ki IP dalen
```

```
def generate_quiz(request):

    # Baqi process wahi rahega jo pehle bataya tha

    response = client.chat(model='llama3.1:8b', messages=[...])

    # ...
```

**Zaroori Baat (Firewall Warning)**

Agar connection nahi ho raha, to AI wale PC ki **Windows Firewall** block kar rahi hogi:

- Control Panel > Windows Defender Firewall > **Advanced Settings** mein jayen.
- **Inbound Rules** mein "New Rule" banayen.
- **Port** select karen aur 11434 likh kar "Allow the connection" kar den.

**Summary Diagram**

| System | Role                       | Address to Use            |
|--------|----------------------------|---------------------------|
| PC 1   | AI (Ollama)                | 192.168.10.20 (Example)   |
| PC 2   | Development (Django/React) | Connecting to PC 1 via IP |

Agar aapka connection successfully test ho gaya hai, to kya mein aapko woh Python script likh kar doon jo automatic PDF read karke AI PC ko data bhejegi?